

Istio — Architecture & Key Terms Explained

Source: KodeKloud — Istio Service Mesh / Istio Introduction / Istio page **What this page teaches:** What Istio is, how it works, and the architecture behind it.

What Is Istio?

Istio is an **open-source service mesh** — think of it as a ready-made, production-grade tool that handles all the networking headaches we talked about earlier (finding services, load balancing, security, monitoring).

Key point: It works with **Kubernetes** AND traditional (non-Kubernetes) apps.

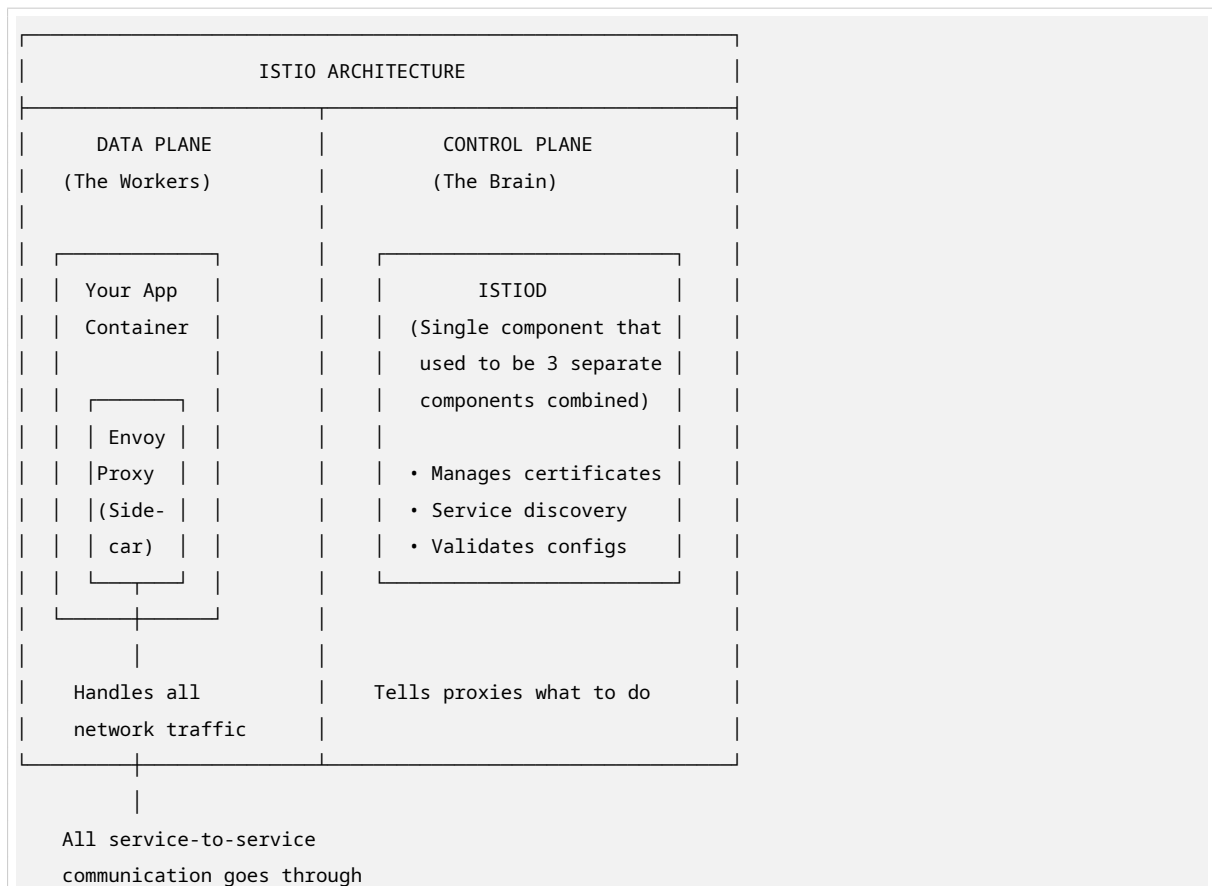
What it gives you:

- **Traffic management** — control how requests flow between services
- **Telemetry** — collect data about what's happening (metrics, logs, traces)
- **Security** — encrypt and authenticate communication between services

It's backed by big cloud providers (Google, IBM, etc.) and is one of the most popular service mesh tools.

Istio Architecture (The Big Picture)

Istio has **two main parts**:



these proxies

Part 1: Data Plane (The Workers)

What is it?

The **Data Plane** is where the actual work happens. It consists of **Envoy proxies** that sit next to your application in every Pod.

What is Envoy?

Envoy is a high-performance proxy (like a smart traffic cop). Istio uses Envoy as its sidecar proxy.

What does the Data Plane do?

- **Load balancing** — spreads traffic evenly
- **Security** — encrypts communication
- **Observability** — collects metrics and logs about every request

In simple terms: The Data Plane is like the **army of waiters** in a restaurant. They take orders, deliver food, check if the kitchen is open, and report back to the manager.

Part 2: Control Plane (The Brain)

The **Control Plane** tells the Data Plane what to do. It configures the proxies, sets routing rules, enforces security policies, and collects data.

The Old Way (Istio v1.x)

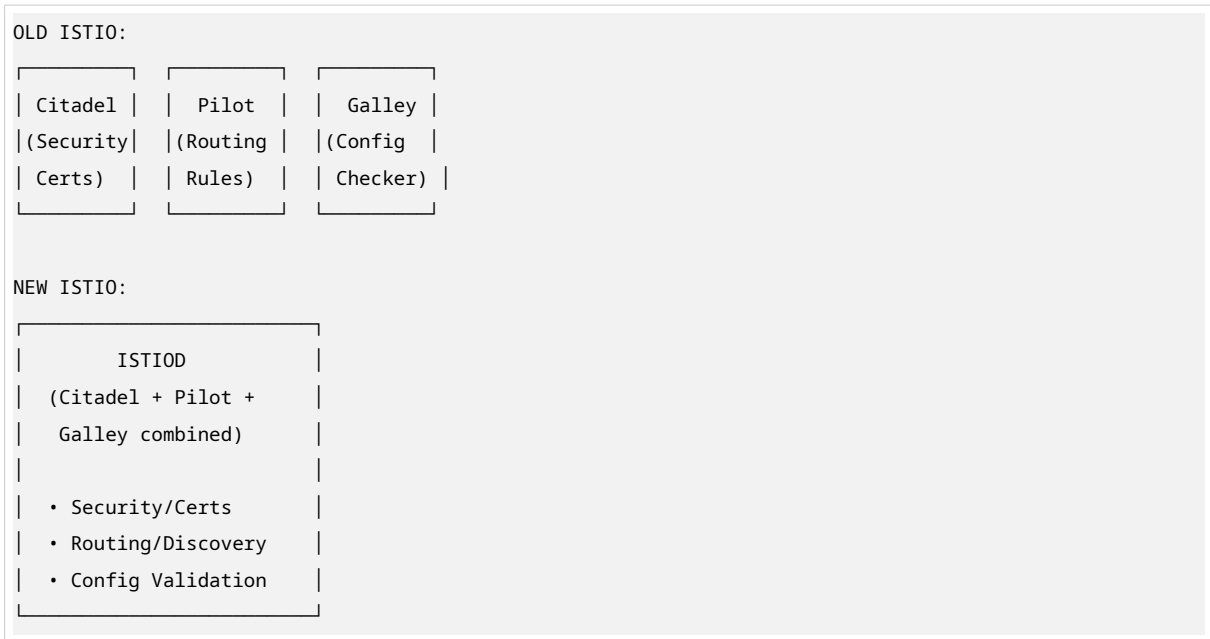
Originally, Istio's Control Plane had **three separate components**:

Component	What It Did	Simple Analogy
Citadel	Generated and managed certificates for secure (encrypted) communication between services	The security guard who issues ID badges to all staff
Pilot	Handled service discovery and kept routing configurations up to date	The manager who knows where everyone works and assigns tasks
Galley	Validated configuration files to make sure settings were correct	The quality checker who reviews the rulebook before it's published

The New Way (Modern Istio)

These three were **consolidated into one single component** called **Istiod**.

Why? Having three separate pieces was complex to manage. Combining them into one makes Istio simpler to install and operate.



The Istio Agent

Inside every Pod, alongside your app and the Envoy proxy, there's an **Istio Agent**.

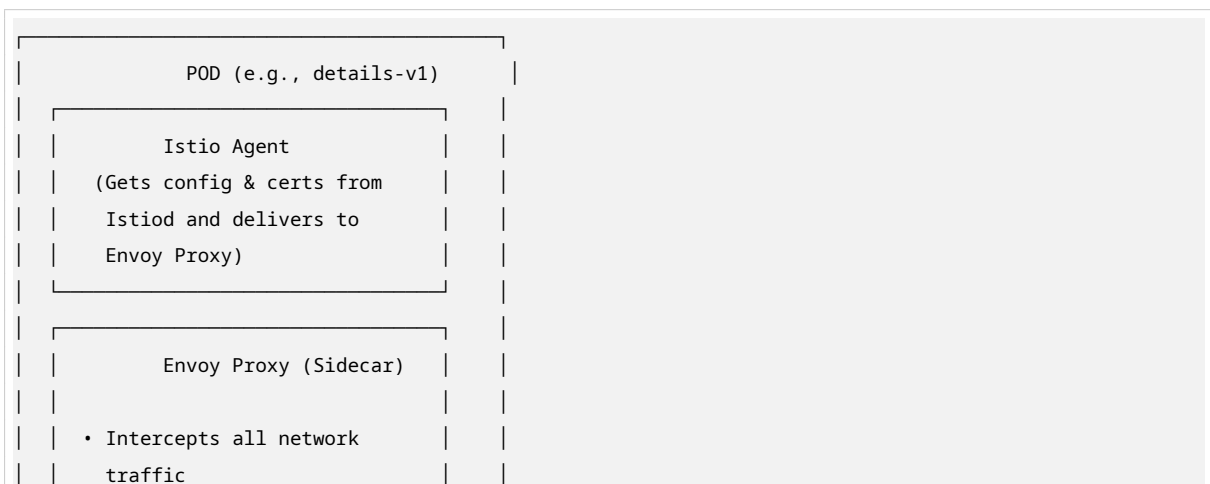
What does the Istio Agent do?

- Delivers **configuration** from Istiod to the Envoy proxy
- Delivers **secrets** (like certificates) to the proxy
- Makes sure the proxy has everything it needs to work correctly

Analogy: If Envoy is the waiter, and Istiod is the restaurant manager, then the **Istio Agent** is the **walkie-talkie** that keeps the waiter updated with the latest menu changes and special instructions.

Full Architecture Visual (Pod Level)

Here's what a single Pod looks like with Istio:

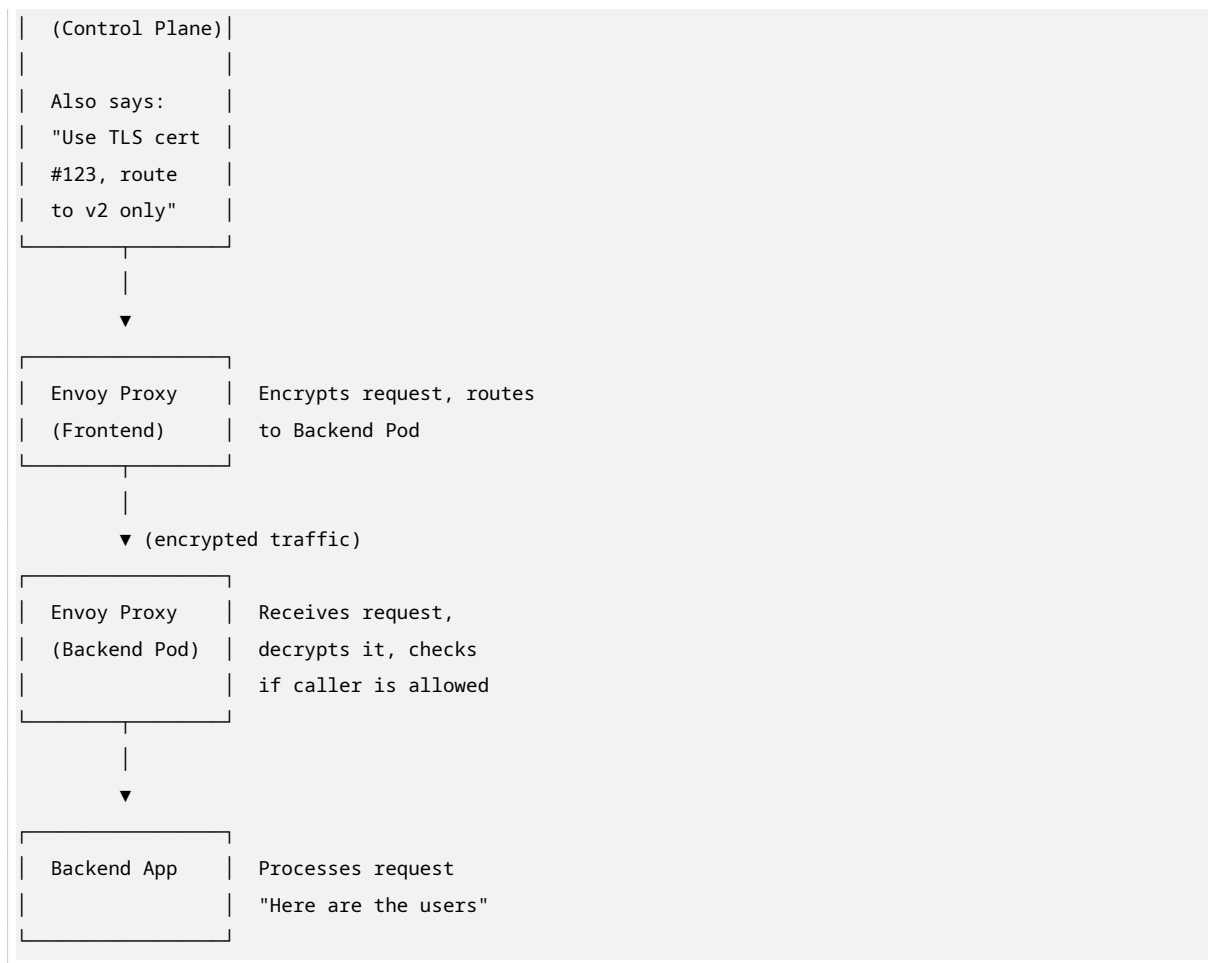




How It All Works Together (The Flow)

Let's trace a request from **Frontend Pod** to **Backend Pod**:





All Terms Explained

Term	Simple Definition
Istio	A popular open-source tool that adds a service mesh to your Kubernetes cluster
Service Mesh	A dedicated infrastructure layer that handles service-to-service communication
Data Plane	The layer that actually moves traffic (Envoy proxies)
Control Plane	The brain that configures and manages the proxies (Istiod)
Envoy	The proxy software that Istio uses as its sidecar
Sidecar	A helper container that runs alongside your app in the same Pod
Citadel	Old Istio component for managing security certificates (now part of Istiod)
Pilot	Old Istio component for service discovery and routing (now part of Istiod)
Galley	Old Istio component for validating configurations (now part of Istiod)

Table 2 – continued

Term	Simple Definition
Istiod	The modern, combined Control Plane (replaces Citadel + Pilot + Galley)
Istio Agent	A small helper in each Pod that feeds config and certificates to Envoy
Telemetry	Data collection –metrics, logs, traces about your traffic
TLS / Certificates	Digital “ID cards” that prove a service is who it says it is, and encrypt traffic
mTLS	Mutual TLS –both sides of a connection verify each other’ s identity
Service Discovery	Automatically finding where services are running (IP addresses, ports)
Load Balancing	Spreading traffic across multiple instances of a service
Observability	The ability to see what’ s happening in your system (metrics, logs, traces)
Ingress Gateway	The entry point for external traffic coming INTO your cluster
Egress Gateway	The exit point for traffic going OUT of your cluster to external services
Namespace	A virtual partition in Kubernetes to group related resources
Pod	The smallest deployable unit in Kubernetes –holds your containers
Deployment	A Kubernetes object that ensures a certain number of Pods are always running
Service	A stable network address that routes traffic to a group of Pods
CustomResourceDefinition (CRD)	A way to extend Kubernetes with new object types
Webhook	A mechanism that intercepts requests to validate or modify them
Mutating Webhook	A webhook that can MODIFY objects (like adding a sidecar) before they are created
ConfigMap	A Kubernetes object that stores configuration data as key-value pairs
Secret	A Kubernetes object that stores sensitive data (passwords, certificates, tokens)
ServiceAccount	An identity assigned to Pods so they can interact with the Kubernetes API
RBAC (Role-Based Access Control)	Permissions system that controls who can do what in Kubernetes
ClusterRole	Permissions that apply across the entire cluster
RoleBinding	Attaches permissions (Roles) to identities (ServiceAccounts)

Why This Matters (Recap)

Without Istio	With Istio
You write networking code in your app	Your app stays clean —Istio handles networking
You manually manage service addresses	Automatic service discovery
No built-in encryption	Automatic mTLS (mutual TLS) encryption
Hard to see what's failing	Full observability dashboard
Complex retry/timeout logic	Built-in resilience patterns
Multiple tools needed	One unified platform

One-Sentence Summary

Istio is a tool that injects a smart proxy (Envoy) into every Pod and manages all those proxies through a central brain (Istiod), so your microservices can talk to each other securely, reliably, and observably — without you writing a single line of networking code.